

datasource_adapter [↗](#)

DefaultParameters module-attribute [↗](#)

```
DefaultParameters = namedtuple('DefaultParameters', ('has_header', 'sep', 'quote', 'line_separat  
or', 'encoding', 'missing_values', 'merge_schemas', 'n_records', 'cat_threshold', 'dtypes'), def  
aults=(True, ',', '"', '\n', name(), {'', 'NA', 'NULL'}, False, DEFAULT_NUMBER_OF_RECORDS, DEFAU  
LT_MAX_CATEGORIES, {}))
```

MS module-attribute [↗](#)

```
MS = 1000
```

The number of milliseconds in one second. Used to convert to/from spark timestamps.

logger module-attribute [↗](#)

```
logger = getLogger(__name__)
```

SparkDatasourceAdapter [↗](#)

```
SparkDatasourceAdapter(spark\_session)
```

Bases: `object`

A class that follows the adapter pattern. This class is responsible for:

- adapting pulse data sources into spark dataframes, through `from_pulse_datasource`
- adapting spark dataframes into pulse data sources, through `to_pulse_datasource`

Parameters:

Name	Type	Description	Default
<code>spark_session</code> ↗	<code>SparkSession</code>	Session to which the adapter is bound to.	<i>required</i>

spark_session instance-attribute [↗](#)

```
spark_session = spark\_session
```

from_pulse_datasource [↗](#)

```
from_pulse_datasource(app, name)
```

Adapts a pulse data source into a pyspark dataframe.

Parameters:

Name	Type	Description	Default
<code>app</code> ↗	<code>App</code>	The app in which the data source exists.	<i>required</i>
	<code>str</code>		<i>required</i>

Name	Type	Description	Default
<code>name</code> ↗		The Data Source name as it appears in the UI.	

Returns:

Type	Description
<code>DataFrame</code>	DataFrame equivalent to the referenced pulse data source.

`to_pulse_datasource` [↗](#)

```
to_pulse_datasource(dataframe: _TDataFrame, app: App, name: str, path: str, file_format: str = 'csv', schema: Optional[Schema] = None, **kwargs: Union[bool, str, int, Dict[str, Any], Set[str]])
```

Adapts a pyspark dataframe into a pulse datasource. The dataframe will be written in the path specified before being added to Pulse. DateType and TimestampType fields are written as unix time (in milliseconds). The schema will be automatically inferred using `AbstractDataSource#inferSchema` approach if the Schema is not specified.

All pySpark types but DecimalType and MLlib types are supported.

Parameters:

Name	Type	Description	Default
<code>dataframe</code> ↗	<code>DataFrame</code>	DataFrame which will be used to create a Pulse datasource.	<i>required</i>
<code>app</code> ↗	<code>App</code>	The App where the datasource will be created.	<i>required</i>
<code>name</code> ↗	<code>str</code>	The name of the datasource.	<i>required</i>
<code>path</code> ↗	<code>str</code>	The path of the file containing the dataset.	<i>required</i>
<code>file_format</code> ↗	<code>str</code>	The input file format. Options are <code>csv</code> and <code>parquet</code> . Defaults to <code>csv</code> .	<code>'csv'</code>
<code>schema</code> ↗	<code>Schema</code>	The Pulse Schema to use. Optional. If not specified, the Schema will be inferred by Pulse. Only used for csv files.	<code>None</code>
<code>kwargs</code> ↗	<code>Dict</code>	Keyword arguments used for the configuration of the DataSource and for the SchemaInference job.	<code>{}</code>

Other Parameters:

Name	Type	Description
<code>has_header</code>	<code>bool</code>	Flags whether the dataset is to be read with header. Only used for csv files. Defaults to True.
<code>sep</code>	<code>str</code>	Delimiter to use. Should contain exactly one character. Only used for csv files. Defaults to <code>','</code> .

Name	Type	Description
missing_values	set of str	Set of strings with values for empty value. Only used for csv files. Defaults to {"", "NA", "NULL"}.
quote	str	The character used to denote the start and end of a quoted item. Only used for csv files. Defaults to "".
line_separator	str	The string used mark the end of line. Only used for csv files. Defaults to "\n".
encoding	str	Encoding to use for UTF when reading/writing. Only used for csv files. Defaults to 'UTF-8'.
cat_threshold	int	A value that represents the maximum number of distinct values on a given field for it to be inferred as categorical. Defaults to 15. Will only be used if no Schema is specified.
job_properties	Dict	The dictionary with the specific properties for the schema inference job. Will only be used if no Schema is specified. Defaults to Pulse PropertiesProfile.
n_records	int	The number of records that will be read per partition when performing datasource schema inference. Will only be used if no Schema is specified. Defaults to 15000.
dtypes	Dict	The dictionary with the field overriding information when inferring the datasource schema. Will only be used for csv files and if no Schema is specified. Defaults to None.
merge_schemas	bool	Specify if schema merging should be allowed. Only used for parquet files. Defaults to False. See: https://spark.apache.org/docs/latest/sql-data-sources-parquet.html#schema-merging

Returns:

Type	Description
DataSource	The new DataSource instance.